

# Support IQ (dSQ) - Production Verification Report

**Report Generated:** January 26, 2026 **Production URL:** <https://support-iq-pearl.vercel.app/dsq> **Test Status:** 125/125 PASSED (100%)

## Executive Summary

Full production verification of Support IQ (dSQ) completed successfully. All semantic pattern matching, vector embeddings, and widget responses are functioning correctly across all 3 modes, 10 personas, and 33 widget types.

| Metric           | Value  | Status |
|------------------|--------|--------|
| Total Tests      | 125    | -      |
| Passed           | 125    | ✓      |
| Failed           | 0      | ✓      |
| Pass Rate        | 100.0% | ✓      |
| Modes Tested     | 3/3    | ✓      |
| Personas Tested  | 10/10  | ✓      |
| Widgets Verified | 33/33  | ✓      |

## Results by Mode

| Mode       | Tests | Passed | Pass Rate | Status |
|------------|-------|--------|-----------|--------|
| Government | 37    | 37     | 100.0%    | ✓      |
| Project    | 34    | 34     | 100.0%    | ✓      |
| ATC        | 54    | 54     | 100.0%    | ✓      |

## Results by Persona

### Government Mode

| Persona          | Role                                 | Tests | Passed | Status |
|------------------|--------------------------------------|-------|--------|--------|
| COR              | Contracting Officer's Representative | 12    | 12     | ✓      |
| Program Manager  | Program Manager                      | 12    | 12     | ✓      |
| Stakeholder Lead | Stakeholder Lead                     | 10    | 10     | ✓      |

### Project Mode

| Persona | Role | Tests | Passed | Status |
|---------|------|-------|--------|--------|
|---------|------|-------|--------|--------|

|                     |                     |    |    |   |
|---------------------|---------------------|----|----|---|
| Project Manager     | Project Manager     | 10 | 10 | ✓ |
| Service Team Lead   | Service Team Lead   | 11 | 11 | ✓ |
| Service Team Member | Service Team Member | 10 | 10 | ✓ |

**ATC Mode (Enterprise Support)**

| Persona       | Role                     | Tests | Passed | Status |
|---------------|--------------------------|-------|--------|--------|
| ATC Executive | C-Level Executive        | 10    | 10     | ✓      |
| ATC Manager   | CS Manager               | 11    | 11     | ✓      |
| ATC Support   | Support Agent            | 17    | 17     | ✓      |
| ATC CSM       | Customer Success Manager | 13    | 13     | ✓      |

**Widget Coverage**

All 33 widget types verified with 100% pass rate:

| Widget                           | Tests | Passed | Status |
|----------------------------------|-------|--------|--------|
| agent-performance-comparison     | 9     | 9      | ✓      |
| customer-risk-list               | 8     | 8      | ✓      |
| team-workload-dashboard          | 7     | 7      | ✓      |
| analytics-dashboard              | 7     | 7      | ✓      |
| program-health-dashboard         | 6     | 6      | ✓      |
| contract-performance-dashboard   | 5     | 5      | ✓      |
| code-quality-dashboard           | 5     | 5      | ✓      |
| agent-dashboard                  | 5     | 5      | ✓      |
| resource-capacity-dashboard      | 4     | 4      | ✓      |
| sprint-burndown-chart            | 4     | 4      | ✓      |
| task-kanban-board                | 4     | 4      | ✓      |
| agent-performance-stats          | 4     | 4      | ✓      |
| vendor-compliance-dashboard      | 3     | 3      | ✓      |
| change-request-dashboard         | 3     | 3      | ✓      |
| stakeholder-engagement-dashboard | 3     | 3      | ✓      |
| sentiment-analysis               | 3     | 3      | ✓      |

|                                 |   |   |   |
|---------------------------------|---|---|---|
| requirements-tracking-dashboard | 3 | 3 | ✓ |
| dora-metrics-dashboard          | 3 | 3 | ✓ |
| ticket-list                     | 3 | 3 | ✓ |
| ticket-detail                   | 3 | 3 | ✓ |
| deliverable-review-list         | 2 | 2 | ✓ |
| meeting-scheduler               | 2 | 2 | ✓ |
| team-velocity-dashboard         | 2 | 2 | ✓ |
| blocker-resolution-dashboard    | 2 | 2 | ✓ |
| deployment-pipeline-dashboard   | 2 | 2 | ✓ |
| executive-summary               | 2 | 2 | ✓ |
| sla-performance-chart           | 2 | 2 | ✓ |
| similar-tickets-analysis        | 2 | 2 | ✓ |
| call-prep-notes                 | 2 | 2 | ✓ |
| response-composer               | 2 | 2 | ✓ |
| knowledge-article               | 2 | 2 | ✓ |
| knowledge-base-search           | 1 | 1 | ✓ |
| customer-risk-profile           | 1 | 1 | ✓ |

## Technical Verification

### Pattern Matching System

- **Algorithm:** Jaccard + Levenshtein fuzzy matching
- **Total Patterns:** 148 semantic patterns
- **Status:** ✓ All patterns matching correctly

### Vector Embeddings (pgvector)

- **Extension:** pgvector v0.8.0
- **Model:** OpenAI text-embedding-3-small
- **Dimensions:** 1536
- **Coverage:** 100% (382 vectors)
- **Status:** ✓ All embeddings operational

### Database Tables with Embeddings

| Table           | Records | Embedding Coverage | Status |
|-----------------|---------|--------------------|--------|
| knowledge_items | 356     | 100%               | ✓      |
| kb_articles     | 10      | 100%               | ✓      |

|                  |            |             |   |
|------------------|------------|-------------|---|
| canned_responses | 8          | 100%        | ✓ |
| tickets          | 8          | 100%        | ✓ |
| <b>Total</b>     | <b>382</b> | <b>100%</b> | ✓ |

## Verified Queries by Persona

### COR - Alexa Johnson (Government)

| Query                                | Widget                         | Status |
|--------------------------------------|--------------------------------|--------|
| Show contract status                 | contract-performance-dashboard | ✓      |
| Show me current contract status      | contract-performance-dashboard | ✓      |
| Show contract performance dashboard  | contract-performance-dashboard | ✓      |
| Show vendor performance              | vendor-compliance-dashboard    | ✓      |
| Show me vendor performance metrics   | vendor-compliance-dashboard    | ✓      |
| Show me compliance dashboard         | vendor-compliance-dashboard    | ✓      |
| Show deliverables due this month     | deliverable-review-list        | ✓      |
| Show me contract deliverables status | deliverable-review-list        | ✓      |
| Show me budget tracking dashboard    | contract-performance-dashboard | ✓      |
| Budget remaining for contracts       | contract-performance-dashboard | ✓      |
| Who is top performing agent?         | agent-performance-comparison   | ✓      |
| Who is most slacking agent?          | team-workload-dashboard        | ✓      |

### Program Manager - Jennifer Chen (Government)

| Query                         | Widget                      | Status |
|-------------------------------|-----------------------------|--------|
| Show program overview         | program-health-dashboard    | ✓      |
| Show program health dashboard | program-health-dashboard    | ✓      |
| Program health                | program-health-dashboard    | ✓      |
| Show milestone status         | program-health-dashboard    | ✓      |
| Milestone status              | program-health-dashboard    | ✓      |
| Show risk register            | program-health-dashboard    | ✓      |
| Critical risk                 | change-request-dashboard    | ✓      |
| Show resource allocation      | resource-capacity-dashboard | ✓      |

|                         |                              |   |
|-------------------------|------------------------------|---|
| Resource capacity       | resource-capacity-dashboard  | ✓ |
| Show me sprint burndown | sprint-burndown-chart        | ✓ |
| Sprint burn-down        | sprint-burndown-chart        | ✓ |
| top performers          | agent-performance-comparison | ✓ |

#### Stakeholder Lead - Jessica Martinez (Government)

| Query                         | Widget                           | Status |
|-------------------------------|----------------------------------|--------|
| Show impact analysis          | stakeholder-engagement-dashboard | ✓      |
| Stakeholder engagement status | stakeholder-engagement-dashboard | ✓      |
| Show stakeholder engagement   | stakeholder-engagement-dashboard | ✓      |
| Show change requests          | change-request-dashboard         | ✓      |
| Change request pending        | change-request-dashboard         | ✓      |
| Show user feedback            | sentiment-analysis               | ✓      |
| Show requirements tracking    | requirements-tracking-dashboard  | ✓      |
| Requirements tracking status  | requirements-tracking-dashboard  | ✓      |
| Requirements traceability     | requirements-tracking-dashboard  | ✓      |
| Upcoming meetings             | meeting-scheduler                | ✓      |

#### Project Manager - Dale Thompson (Project)

| Query                  | Widget                       | Status |
|------------------------|------------------------------|--------|
| Show sprint burndown   | sprint-burndown-chart        | ✓      |
| Burndown               | sprint-burndown-chart        | ✓      |
| Show team velocity     | team-velocity-dashboard      | ✓      |
| Velocity               | team-velocity-dashboard      | ✓      |
| Show resource capacity | resource-capacity-dashboard  | ✓      |
| Resource capacity      | resource-capacity-dashboard  | ✓      |
| Show blockers          | blocker-resolution-dashboard | ✓      |
| Blocker                | blocker-resolution-dashboard | ✓      |
| Sprint planning        | task-kanban-board            | ✓      |
| top performers         | agent-performance-comparison | ✓      |

#### Service Team Lead - Herbert Roberts (Project)

| Query                     | Widget                        | Status |
|---------------------------|-------------------------------|--------|
| Show team workload        | team-workload-dashboard       | ✓      |
| Team workload             | team-workload-dashboard       | ✓      |
| Show code quality metrics | code-quality-dashboard        | ✓      |
| Code quality              | code-quality-dashboard        | ✓      |
| technical debt            | code-quality-dashboard        | ✓      |
| Show code reviews         | code-quality-dashboard        | ✓      |
| Show deployment status    | deployment-pipeline-dashboard | ✓      |
| Deployment                | deployment-pipeline-dashboard | ✓      |
| DORA metrics              | dora-metrics-dashboard        | ✓      |
| DORA                      | dora-metrics-dashboard        | ✓      |
| Show DORA metrics         | dora-metrics-dashboard        | ✓      |

**Service Team Member - Molly Rivera (Project)**

| Query                         | Widget                       | Status |
|-------------------------------|------------------------------|--------|
| Show my assigned requests     | agent-dashboard              | ✓      |
| Daily update                  | agent-dashboard              | ✓      |
| my dashboard                  | agent-performance-stats      | ✓      |
| Show my performance this week | agent-performance-stats      | ✓      |
| my performance                | agent-performance-stats      | ✓      |
| Show my sprint tasks          | task-kanban-board            | ✓      |
| My tasks                      | task-kanban-board            | ✓      |
| Sprint task                   | task-kanban-board            | ✓      |
| code quality                  | code-quality-dashboard       | ✓      |
| top performers                | agent-performance-comparison | ✓      |

**C-Level Executive - Jennifer Anderson (ATC)**

| Query                     | Widget              | Status |
|---------------------------|---------------------|--------|
| Show me executive summary | executive-summary   | ✓      |
| Show board-level metrics  | executive-summary   | ✓      |
| Show detailed analytics   | analytics-dashboard | ✓      |

|                                       |                              |   |
|---------------------------------------|------------------------------|---|
| Show me the detailed analytics        | analytics-dashboard          | ✓ |
| Show me SLA performance               | sla-performance-chart        | ✓ |
| Show me the SLA performance breakdown | sla-performance-chart        | ✓ |
| Which customers are at churn risk?    | customer-risk-list           | ✓ |
| Show me high-risk customers           | customer-risk-list           | ✓ |
| customer sentiment                    | sentiment-analysis           | ✓ |
| top performers                        | agent-performance-comparison | ✓ |

### CS Manager - David Miller (ATC)

| Query                                  | Widget                       | Status |
|----------------------------------------|------------------------------|--------|
| Show me team status                    | team-workload-dashboard      | ✓      |
| Show me my team's status               | team-workload-dashboard      | ✓      |
| Show workload balance                  | team-workload-dashboard      | ✓      |
| Who is top performing agent?           | agent-performance-comparison | ✓      |
| Who are the top and bottom performers? | agent-performance-comparison | ✓      |
| compare agent performance              | agent-performance-comparison | ✓      |
| Who is most slacking agent?            | team-workload-dashboard      | ✓      |
| Show me all high-risk customers        | customer-risk-list           | ✓      |
| Show team budget                       | analytics-dashboard          | ✓      |
| my current tickets                     | ticket-list                  | ✓      |
| Show ticket DESK-1001                  | ticket-detail                | ✓      |

### Support Agent - Christopher Hayes (ATC)

| Query                                   | Widget                  | Status |
|-----------------------------------------|-------------------------|--------|
| Good morning, what's on my plate today? | agent-dashboard         | ✓      |
| what is on my plate today               | agent-dashboard         | ✓      |
| good morning                            | agent-dashboard         | ✓      |
| Show me my performance stats            | agent-performance-stats | ✓      |
| Show me my tickets                      | ticket-list             | ✓      |
| my tickets                              | ticket-list             | ✓      |
| Show ticket DESK-1001                   | ticket-detail           | ✓      |

|                                             |                          |   |
|---------------------------------------------|--------------------------|---|
| Show me ticket TICK-001                     | ticket-detail            | ✓ |
| Find similar tickets I've resolved          | similar-tickets-analysis | ✓ |
| similar tickets                             | similar-tickets-analysis | ✓ |
| Help me prepare for the call with Acme Corp | call-prep-notes          | ✓ |
| prepare for call                            | call-prep-notes          | ✓ |
| Draft response for angry customer           | response-composer        | ✓ |
| draft response                              | response-composer        | ✓ |
| Search knowledge base for password reset    | knowledge-article        | ✓ |
| knowledge base                              | knowledge-base-search    | ✓ |
| password reset                              | knowledge-article        | ✓ |

### CSM - Jordan Taylor (ATC)

| Query                               | Widget                       | Status |
|-------------------------------------|------------------------------|--------|
| Show customer health scores         | customer-risk-list           | ✓      |
| customer health scores              | customer-risk-list           | ✓      |
| Show upcoming renewals              | analytics-dashboard          | ✓      |
| upcoming renewals                   | analytics-dashboard          | ✓      |
| Show upsell opportunities           | customer-risk-list           | ✓      |
| upsell opportunities                | customer-risk-list           | ✓      |
| expansion opportunities             | customer-risk-list           | ✓      |
| Which customers declining adoption? | analytics-dashboard          | ✓      |
| product adoption                    | analytics-dashboard          | ✓      |
| churn risk analysis                 | customer-risk-profile        | ✓      |
| NPS survey results                  | sentiment-analysis           | ✓      |
| business review                     | meeting-scheduler            | ✓      |
| top performers                      | agent-performance-comparison | ✓      |

### Universal Queries (Cross-Mode)

| Query                   | Tested Personas                   | Widget      | Status |
|-------------------------|-----------------------------------|-------------|--------|
| Show me zoho tickets    | cor, project-manager, atc-support | ticket-list | ✓      |
| show me my zoho tickets | cor, project-manager, atc-support | ticket-list | ✓      |



|                 |                                   |             |                                                                                     |
|-----------------|-----------------------------------|-------------|-------------------------------------------------------------------------------------|
| Show me tickets | cor, project-manager, atc-support | ticket-list |  |
|-----------------|-----------------------------------|-------------|-------------------------------------------------------------------------------------|

## Conclusion

**VERIFICATION STATUS:**  **PASSED**

All 125 test cases passed with 100% success rate. The Support IQ (dSQ) production system is fully operational with:

1. **Pattern Matching:** All 148 semantic patterns functioning correctly
2. **Vector Embeddings:** 382 vectors with 100% coverage using pgvector v0.8.0
3. **Widget Responses:** All 33 widget types returning correct responses
4. **Mode Coverage:** All 3 modes (Government, Project, ATC) verified
5. **Persona Coverage:** All 10 personas verified
6. **Cross-Mode Queries:** Universal queries working across all personas

**Report Generated By:** Automated Test Suite **Test Duration:** ~35 seconds **Production Environment:** Vercel (support-iq-pearl.vercel.app)